

A Decision-Evaluation Benchmark for EHR World Models and Offline Reinforcement Learning

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ABSTRACT

Electronic health record (EHR) world models are increasingly connected to offline reinforcement learning (RL), yet comparisons can be driven by incompatible tasks, action support, rewards, or evaluator assumptions rather than algorithmic differences. We introduce a development benchmark that standardizes these layers across six ICU task constructions and evaluates six transition methods, five uncertainty-capable methods, 34 fixed policy/planner labels, and nine off-policy evaluation (OPE) estimators. No method is a global winner: the Transformer has the lowest one-step error on all six tasks, whereas final-horizon error is led by the Gaussian ensemble on five and persistence/LOCF on one; recursive CRPS is led by the ensemble on all six, while recursive calibration error is led by GRU-D on all six. Known-value policy comparison covers respiratory, shock, AKI-RRT, and heart-failure extensions. In moderate regimes, fixed extreme-action controls lead all four tasks—maximum supported action for respiratory, shock, and heart failure, but minimum supported action for AKI—while world-model planners are tied or narrowly separated. OPE authorization is task-specific: a shared aggregate-calibrated evaluator approves 0 of 3,456 tuples, an AKI-matched evaluator approves 236 of 1,728, and a heart-failure-matched evaluator approves 0 of 2,880. Even the AKI tuples do not authorize retrospective scoring because trained target-policy and post-training overlap gates are incomplete. We report no retrospective EHR policy value or winner. The contribution is an executable technical scaffold and failure decomposition from forecasting through policy-evaluation authorization.

CCS CONCEPTS

• **Applied computing** → **Health informatics**; • **Computing methodologies** → **Reinforcement learning**.

KEYWORDS

electronic health records, world models, offline reinforcement learning, benchmark, off-policy evaluation, uncertainty

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1 INTRODUCTION

EHR prediction benchmarks standardize cohorts, preprocessing, splits, and metrics [4, 12]. Offline RL over EHR trajectories adds a more fragile technical chain: recorded state must precede the action label, reward must follow it without mechanically encoding it,

policy probability must remain inside empirical support, learned dynamics must remain stable under recursion, and off-policy evaluation (OPE) must recover known values before retrospective scoring is authorized. Sepsis RL established a widely reused fluid-by-vasopressor action abstraction [6]; recent work connects irregular EHR world models and latent imagination to policy learning [13]. These studies motivate a shared evaluator, but their different task and scoring contracts prevent direct algorithmic comparison.

Our primary contribution is the standardized comparison and failure decomposition across EHR world-model learning, world model $\times$ planner combinations, model-based and model-free offline RL, known-value truth, retrospective support diagnostics, and cross-cohort evaluability. Transition accuracy is a validity layer rather than the endpoint. Temporal leakage control is a methods requirement rather than the headline contribution.

Because no genuinely untouched role remains, all EHR findings are labeled *development benchmark evidence*. The benchmark evaluates world-model-based and model-free offline-RL pipelines across heterogeneous EHR tasks, compares true policy performance in matched known-value environments, and audits support, collapse, planning stability, and model exploitation on retrospective EHR data. The retrospective layer evaluates computational behavior, not policy value.

Contributions. We provide: (1) a common, time-ordered task scaffold with train-only fitting and subject-disjoint development roles; (2) a canonical sepsis published-compatibility reference plus five task-specific extensions; (3) four task-matched known-value policy comparisons under a shared algorithm inventory; (4) a component-level bridge from prediction and uncertainty to true policy performance and model exploitation; (5) complete tuple-level OPE metrics and authorization decisions rather than estimator names alone; and (6) explicit non-execution when probability, support, or evaluator prerequisites fail.

Research questions. The benchmark asks: **RQ1**, how do world-model families differ in one-step, recursive, reward, termination, uncertainty, and action-information fidelity? **RQ2**, under known action-response mechanisms, how do world-model-planner systems compare with model-free methods and fixed controls? **RQ3**, when do learned models exhibit exploitation gaps, and which OPE contracts recover true policy values and ranks? **RQ4**, which parts of this decision scaffold remain operational across heterogeneous cohorts without transferring sepsis-specific reward semantics?

2 RELATED DECISION BENCHMARKS

Clinical time-series benchmarks standardize prediction tasks but generally stop before action, reward, planning, and policy evaluation [4, 12]. Synthetic medical RL benchmarks such as EpiCare

and DTR-Bench provide controlled treatment-response mechanisms and true returns [3, 10], but they do not test whether a retrospective EHR cohort has temporally valid actions, observable rewards, or sufficient support. Conversely, task-specific MIMIC studies make these choices but are difficult to compare. DeepVent uses four-hour mechanical-ventilation decisions over tidal volume, PEEP, and FiO₂ with an APACHE-style dense reward [7]; POPCORN studies hourly fluid-vasopressor decisions in acute hypotension with a partially observed model and support-aware OPE [2]. Our respiratory and shock tasks are published-inspired adapters, not exact reproductions of either study.

Recent EHR world-model work links irregular observation modeling and latent imagination to policy learning [13]. More broadly, healthcare offline-RL model selection is sensitive to reward, support, and estimator choice [9, 11]. We therefore join real-EHR task construction to known-value evaluation, while withholding retrospective policy rankings when the bridge does not validate.

3 BENCHMARK SCAFFOLD

3.1 State, action, target, and roles

Each transition uses a fixed four-hour clock. Let

$$x_t = [v_t, m_t, \delta_t, c, a_{t-1}], \quad y_t = v_{t+1}, \quad (1)$$

where v_t is the pre-action value vector, m_t its observation mask, δ_t feature-wise recency, c static/pre-anchor context, and a_{t-1} the previous action abstraction. Current action, post-action measurements, future masks, post-outcome fields, and whole-stay diagnosis proxies are prohibited from x_t . The logged action a_t summarizes recorded exposure or settings over $[t, t + 4h)$; y_t begins at $t + 4h$. Every evaluated bin remains inside one ICU stay.

Throughout, $K = |\mathcal{A}|$ denotes the number of discrete action classes, not the number of observed transitions or treatment dimensions.

The interval label is deliberately called a *recorded-exposure action*: it can include adaptation during the interval and is not asserted to be fully assignable at its start. It is valid for logged-action-conditioned forecasting and retrospective policy-behavior diagnostics. In the known-value environments, by contrast, a_t is an assignable intervention in the constructed transition kernel. This distinction prevents the real-EHR label from inheriting the stronger semantics of the simulator action.

Imputation, normalization, action cutpoints, support masks, feature selection, behavior denominators, hyperparameters, and checkpoints are fitted or selected on development train/validation roles only. The roles are subject disjoint, but historical use of the available evaluation roles precludes calling any role a new confirmatory holdout. No row is used to claim a confirmatory best method.

3.2 Canonical sepsis reference

Sepsis uses a published-compatible four-hour scaffold with a training-derived fluid quintile $\times$ vasopressor quintile action ($K = 25$), a paper-listed 40-variable reference view, masks/time gaps, prior action, and local SOFA/lactate reward parameterization. The reference is not an exact reproduction: numerical reward coefficients and parts of the original cohort implementation are

not source-verifiable; the repaired anchor and stay-bounding rules intentionally differ; the local SOFA proxy can mechanically overlap vasopressor action; and anchor-relative 90-day outcome ascertainment is unavailable. We retain a compatibility track and a treatment-isolated reward sensitivity rather than treating compatibility as permission to ignore circularity.

The matched feature/reward/denominator comparison did not support choosing a preferred setting because terminal outcome availability and matched semi-synthetic reliability were incomplete. Sepsis remains the canonical reference despite weak factual action information.

3.3 Task-specific extensions

Table 1 summarizes extension roles and aggregate scale. Subject counts are task-specific development materializations and should not be summed because one subject may satisfy more than one cohort definition. A known-value scenario is one unique response-regime $\times$ reward-variant environment contract; method identities and random seeds are excluded from this count. Respiratory uses observed PEEP quintile $\times$ observed FiO₂ quintile ($K = 25$); a row is excluded when either setting is unavailable rather than mapped to no action. Its primary dense benchmark reward is

$$r_t^{\text{resp}} = \mathbb{I}[94 \leq \text{SpO}_{2,t+1} \leq 98] - .5\mathbb{I}[\text{SpO}_2 \text{ outside}] + \mathbb{I}[70 \leq \text{MBP}_{t+1} \leq 80] - .5\mathbb{I}[\text{MBP outside}], \quad (2)$$

with an SpO₂-only sensitivity. Shock uses fluid quintile $\times$ vasopressor quintile and two separated reward components,

$$r_t^{\text{MAP}} = \text{clip}((\text{MBP}_{t+1} - 65)/25, -1, 1), \quad r_t^{\text{lac}} = -\tanh(\text{lac}_{t+1} - \text{lac}_t). \quad (3)$$

All are frozen post-action benchmark objectives rather than transferred sepsis semantics. Sepsis world-model reward diagnostics use the lactate component; its reference terminal outcome is unavailable. AF/flutter remains world-model-only because strict action-reward ordering is unresolved. AKI and heart failure retain their original P/T transition surfaces and add separate decision-task extensions: severe AKI uses daily wait versus initiate RRT ($K = 2$, absorbing after initiation, at most three decisions), while heart failure uses a six-hour loop-diuretic-quartile by vaso/inotrope indicator action ($K = 8$, at most 12 decisions). Their post-action reward proxies define task-matched known-value mechanisms; they do not create retrospective EHR policy values.

4 METHODS COMPARED

4.1 World models and component metrics

The evaluated component is

$$\hat{M} = (\hat{P}_\theta(v_{t+1} | x_t, a_t), \hat{R}, \hat{T}), \quad (4)$$

not a complete generator of the future observation process. The core includes recurrent/GRU-D, causal Transformer, categorical recurrent state-space, and Gaussian recurrent ensemble models; persistence and histogram gradient boosting are sanity controls. Neural runs use seeds 3408/3411/3414, hidden dimension 48, latent dimension 16 where applicable, batch size 512, learning rate 0.002, and weight decay 10^{-4} . Checkpoints minimize 0.7 RMSE+0.3 MAE

Table 1: Development task scale and evaluation layers. “Supported/ K ” gives train-supported EHR action classes over the total discrete action space. Known-value scenarios count unique response-regime $\times$ reward-variant contracts, excluding methods and seeds. AKI and heart failure additionally use the explicitly named decision extensions below. No row has retrospective EHR policy value.

Cohort	Subjects	KV scenarios	Action contract	Benchmark role	Reward / value status
Sepsis	3,391	0	22/25 fluid-vasopressor	Canonical reference	P/R/T plus support/collapse diagnostics
Respiratory support	18,709	8	12/25 PEEP-FiO ₂	Policy comparison	P/R/T plus known-value matrix
Shock	37,015	8	25/25 fluid-vasopressor	Policy comparison	P/R/T plus known-value matrix
AF/flutter	2,551	0	16/25 rate-rhythm	World-model only	P/T; action-reward ordering unresolved
AKI	16,380	4	WM: 4/4 compact; decision: 2 wait/RRT	Decision extension	P/T plus task-matched known-value matrix
Heart failure	2,378	4	WM: 2/2 compact; decision: 8 diuretic $\times$ vaso/inotrope	Decision extension	P/T plus task-matched known-value matrix

on development validation after train-only fitting, with 8 minimum and 40 maximum epochs, patience 6, and one uniform 80-epoch extension when the frozen convergence trigger fires. The Gaussian ensemble combines the three recurrent members and is one derived row per task.

One-step RMSE/MAE uses only observed target cells after train-fitted normalization. Recursive rollout starts from the first validation state, feeds each predicted value back, marks generated future values as unobserved, increments recency, and follows the logged action sequence for eleven four-hour steps. At each horizon, error is scored against cells observed at that step in the recorded target mask. The scalar recursive error is the normalized observed-cell RMSE at the final executed horizon (step 11, 44 hours), averaged across available executed-seed rows for a method-cohort pair; all 924 horizon-resolved rows are released. Thus recursion tests value-state compounding under a fixed missingness convention; it does not claim to generate future measurement decisions. Reward heads are deterministic mappings from predicted state when available. Termination is evaluated natively, after train-only recalibration, and against a train empirical-hazard baseline.

Uncertainty reporting includes Gaussian NLL/CRPS, coverage at 50, 80, 90, and 95%, interval width and score, and risk-coverage. Factual action information uses one fixed primary estimand throughout: shuffled-action minus observed-action RMSE on reward-relevant targets within matched strata, averaged over the four core model families and executed seeds. Each model-seed contrast also carries a 500-replicate materialized-episode bootstrap interval. It is a predictive-information diagnostic.

4.2 Planners and policies

The planner matrix uses exact supported-action search at horizon one and categorical cross-entropy method (CEM) planning at nominal horizons four and eight. Each CEM iteration evaluates 64 sequences, retains eight elites, updates categorical probabilities with smoothing 0.2, applies the frozen support mask at every step, and executes only the first action before replanning. Three iterations and matched model-query budgets are used. The effective horizon is truncated by the remaining task horizon; therefore AKI’s nominal H4 and H8 variants are both H3-effective and are retained only as budget-contract diagnostics. The uncertainty sensitivity subtracts 0.1 times predictive scale.

The model-free core includes behavior cloning, independent implementations of discrete BCQ [1] and CQL [8], a Soft-SPIBB conceptual adapter, a Decision Transformer contract adapter, and supported random, minimum, maximum, and severity-rule controls. Neural runs use three fixed seeds, one train/validation trial, at most 20,000 training windows, 12 epochs, hidden dimension 64, batch size 512, and discount 0.99. Fidelity labels distinguish independent reimplementations from conceptual adapters. Deterministic controls are prespecified ϵ -soft distributions over the supported set with $\epsilon = 0.001$ for retrospective probability diagnostics; they are not scored by retrospective OPE.

4.3 Known-value and OPE evaluation

Known-value families include small exact finite environments and aggregate-calibrated semi-synthetic environments. The respiratory/shock reference family uses 13 states, 25 actions, 11 decisions, task-specific censoring/missingness rates, and frozen supported-action sets. For action coordinates $(a_1, a_2) \in \{0, \dots, 4\}^2$, define $u(a) = ((a_1 + a_2) - 4)/4$. The probability of improving one severity state is $\text{clip}(0.14 + \lambda u(a), 0.01, 0.35)$ and worsening is $\text{clip}(0.12 - \lambda u(a), 0.01, 0.35)$, with $\lambda \in \{0, 0.012, 0.035, 0.030\}$ for null, weak, moderate, and delayed regimes; the delayed effect is zero for the first four decisions. Dense reward is $0.025(s_t - s_{t+1})$ every third step plus a terminal severity term; terminal-only is a separate sensitivity. This mechanism is intentionally auditable and monotone rather than designed to favor a particular learned method.

Task-matched families preserve the four response regimes and exact versus semi-synthetic tiers while changing action count, horizon, state size, support, and aggregate missingness. AKI uses $K = 2$, three daily decisions, and 5/9 exact/semi-synthetic states; heart failure uses $K = 8$, twelve six-hour decisions, and 8/15 states. All action responses, rewards, and termination mechanisms are known by construction. World-model architectures and policy methods are retrained within each environment; EHR-trained weights are not evaluated in these simulators.

All policies share initial states and exogenous noise streams, enabling paired differences. The reported behavior-relative quantity is the Monte Carlo mean of $G^\pi - G^{\pi_b}$ under common environment randomness; stochastic action streams are independently seeded and recorded. Consequently, an empirical-behavior policy can have a small finite-sample paired difference even though its exact value equals the behavior reference. Each result averages three evaluation seeds and reports the paired Monte Carlo standard

error. The released AKI/HF matrix also retains exact dynamic-programming value differences in separate columns; these are not mixed with the Monte Carlo standard errors used in the main table. Exact finite regret uses dynamic programming; semi-synthetic evaluation reports true policy returns without calling a high-budget planner an oracle.

The OPE matrix tests trajectory IS and WIS; per-decision IS, WPDIS, and CWPDIS; DR and WDR; FQE; and support-restricted variants. A contract fixes estimator, behavior denominator, clipping, support rule, horizon, and regime. Here “coverage” is a policy-set diagnostic within a fixed logged-data realization: the fraction of policy-specific bootstrap intervals containing each policy’s known value. It is not repeated-dataset frequentist coverage. Approval requires this diagnostic ≥ 0.90 ; Spearman rank and pairwise-order recovery ≥ 0.80 ; behavior-relative sign recovery ≥ 0.90 ; false-improvement rate ≤ 0.05 ; median ESS $\geq \max(100, 0.05N)$; zero unsupported target mass; maximum paired true-return SE ≤ 0.03 ; and finite required values. No threshold is changed after inspection.

5 RESULTS

5.1 RQ1: World-model fidelity

The full transition inventory comprises persistence/LOCF and HGB controls plus four world-model families. The Transformer has the lowest one-step RMSE on all six cohorts. At the final recursive horizon, the Gaussian ensemble leads five cohorts, while persistence/LOCF leads heart failure; when restricted to the four learned world-model families, the ensemble leads all six. Across the 24 core world-model-cohort bridge rows, normalized one-step error ranges from 0.669 to 0.773 and final-horizon error from 0.901 to 3.268. Thus the one-step selection reverses on every task. These are descriptive point leaders: neural methods have three seeds, the derived ensemble has one task-level row, and no subject-cluster leader interval is available.

Table 2 consolidates every executed comparison layer. It shows a stronger conclusion than a single aggregate leaderboard: some methods dominate one frozen metric, but the identity of the leader changes across point prediction, probabilistic recursion, policy comparison, and evaluator authorization. The complete 54-row layer-specific method inventory and every granular result row are released.

Under the fixed action-information estimand, cohort means range from 0.0001 to 0.0023 (Table 3). Episode-cluster intervals exclude zero in all 10 model-seed rows for respiratory, shock, and AKI; 2/10 for AF/flutter; and 0/10 for sepsis and heart failure. Statistical separation can coexist with negligible magnitude and does not repair a missing repeated-action contract. The empirical-hazard termination baseline has lower Brier score than native and recalibrated learned heads in all 18 comparable canonical-sepsis rows, so learned termination is a sensitivity rather than an unqualified planning component.

Uncertainty is also cohort- and metric-dependent. Across HGB and the four probabilistic world models, the one-step CRPS leader is the Transformer in five cohorts and the ensemble in sepsis. Under recursive rollout, the ensemble has the lowest CRPS in all six, whereas GRU-D has the lowest MACE in all six. The recursive

90% coverage range is especially broad in sepsis (0.623–0.941) and heart failure (0.657–0.934). Hence neither one-step point error nor one-step uncertainty identifies recursive sharpness and calibration jointly.

5.2 RQ2: Known-value policy performance

Each known-value task contains 34 fixed policy/planner labels and no support-mask bypass. For respiratory and shock, the main figure is balanced and policy-level: all five model-free methods, the same support-constrained H4 planner for each of four world models, and one maximum-supported control. Figure 1 reports each fixed row rather than averaging a 24-row planner factorial against five model-free methods. The release contains all 1,632 respiratory/shock rows across H1/H4/H8, reward, response-strength, and uncertainty-penalty conditions, plus all 816 AKI/HF task-extension rows.

In the moderate primary regimes, the five model-free paired behavior-relative differences range from -0.0008 to 0.0061 for respiratory and -0.0133 to 0.0009 for shock. All four support-constrained H4 world-model rows tie exactly at 0.0366 (mean paired SE 0.0029) and 0.0324 (0.0028), respectively. The fixed maximum-supported control is higher at 0.0673 (0.0036) and 0.0760 (0.0037). The task-matched extensions change the direction and scale: minimum supported action leads AKI at 0.1709 (0.0096), whereas maximum supported action leads heart failure at 0.0243 (0.0027). Table 4 shows that world-model planners remain narrowly separated. These are evaluator stress findings under known mechanisms, not a cross-task RL winner.

5.3 RQ3: Exploitation and OPE disposition

The canonical sepsis surface passes 98 of 1,800 horizon-specific pre-estimator gates. Material seed instability occurs in 26 of 50 rows, unsupported probability mass in 9 of 150, and marginal single-action collapse in 18 of 150. These failures explain why probability-complete training output is not sufficient for policy-value scoring.

World-model planners also exhibit substantial simulator-to-truth disagreement under the injected mechanisms: the mean absolute learned-model-versus-true-return gap is 1.079 across 72 respiratory planner rows and 2.344 across 72 shock rows. These values are on task-specific synthetic reward scales and are not compared across cohorts; they show why learned-model value cannot substitute for true known-value evaluation.

The shared reference evaluator approves 32 of 3,456 exact-finite tuples. All 32 are masked-support, moderate-regime, horizon-one contracts and span CWPDIS, DR, IS, PDIS, WDR, WIS, WPDIS, and support-restricted WPDIS. Table 5 shows why family-name transport is invalid: among 3,456 aggregate-calibrated semi-synthetic tuples, none passes coverage, rank, or sign recovery, even though 3,264 have finite outputs and 1,728 pass support and precision individually.

Task matching changes the result rather than uniformly rescuing OPE. For the separate AKI evaluator, 236 of 1,728 semi-synthetic tuples pass all nine gates across weak, moderate, and delayed regimes and horizons one through three. For heart failure, 0 of 2,880 pass. The AKI result validates only those exact estimator-denominator-clip-support-horizon-regime tuples in

Table 2: End-to-end baseline and method atlas. CRPS and mean absolute coverage error (MACE) are minimized. A scenario is one response-regime $\times$ reward-variant contract; rows also vary method and seed. Point leaders are descriptive within a frozen layer and do not define a global cross-layer winner.

Layer	Metric	Evaluated inventory	Scope	Point-leader pattern
Transition	One-step RMSE	LOCF, HGB, GRU-D, Transformer, RSSM, ensemble (6)	6 EHR cohorts	Transformer 6/6
Transition	Final-horizon RMSE	Same six methods	6 cohorts; step 11 (44 h)	Ensemble 5/6; LOCF 1/6
Uncertainty	One-step CRPS / MACE	HGB plus four probabilistic WMs (5); LOCF unavailable	6 EHR cohorts	CRPS: Transformer 5/6, ensemble 1/6; MACE: Transformer 4/6, GRU-D 2/6
Uncertainty	Recursive CRPS / MACE	Same five probabilistic methods	6 EHR cohorts	CRPS: ensemble 6/6; MACE: GRU-D 6/6
Policy	Paired true Δ_b	5 controls + 5 model-free + 24 WM-planner labels (34)	24 scenarios; 2,448 rows	Maximum-supported control 3/4; minimum-supported control 1/4
OPE	All-nine authorization	IS, WIS, PDIS, WPDIS, CWPDIS, DR, WDR, FQE, support-restricted WPDIS (9)	16,128 contract tuples	32/3,456 shared exact; 228/1,728 AKI exact; 236/1,728 AKI semi; 0/9,216 other; 0 EHR

Table 3: Cohort-specific uncertainty after averaging executed seeds within method and cohort. Parentheses give the minimized score. Cov90 is the range across five probabilistic methods. Δ_a is matched-shuffle minus observed-action RMSE on reward-relevant targets and is predictive, not causal.

Cohort	Δ_a	Lowest one-step CRPS	Lowest recursive CRPS	Lowest recursive MACE	Recursive Cov90 range
Sepsis	0.0015	Ensemble (0.334)	Ensemble (0.452)	GRU-D (0.016)	[0.623, 0.941]
Respiratory support	0.0023	Transformer (0.344)	Ensemble (0.458)	GRU-D (0.030)	[0.768, 0.935]
Shock	0.0022	Transformer (0.347)	Ensemble (0.470)	GRU-D (0.015)	[0.765, 0.926]
AF/flutter	0.0008	Transformer (0.340)	Ensemble (0.458)	GRU-D (0.029)	[0.716, 0.941]
AKI	0.0008	Transformer (0.327)	Ensemble (0.458)	GRU-D (0.023)	[0.728, 0.943]
Heart failure	0.0001	Transformer (0.330)	Ensemble (0.450)	GRU-D (0.025)	[0.657, 0.934]

Table 4: Moderate-regime known-value policy summary. Δ_b is the paired behavior-relative return difference. Ranges span the five model-free methods or all 24 world-model-planner rows. OPE counts are exact tuple approvals within the indicated semi-synthetic evaluator; respiratory/shock share the reference evaluator.

Task	Point leader	Leader Δ_b	Model-free range	WM-planner range	Tier-2 OPE
Respiratory	Maximum supported	0.0673	[−0.0008, 0.0061]	[0.0338, 0.0404]	0/3,456 shared
Shock	Maximum supported	0.0760	[−0.0133, 0.0009]	[0.0286, 0.0392]	0/3,456 shared
AKI-RRT	Minimum supported	0.1709	[0.0053, 0.1416]	[0.1102, 0.1124]	236/1,728 task-matched
Heart failure	Maximum supported	0.0243	[−0.0086, 0.0009]	[0.0130, 0.0166]	0/2,880 task-matched

Table 5: OPE gate pass counts; each tier has 3,456 exact estimator-denominator-clip-support-horizon-regime tuples. Thresholds are frozen in Methods.

Gate	Exact finite	Aggregate-calibrated
Coverage	108	0
Rank	144	0
Pairwise order	800	648
Improvement sign	137	0
False improvement	1,204	215
ESS	279	270
Support	1,728	1,728
Paired precision	1,728	1,728
Finite values	3,264	3,264
All nine	32	0

the constructed environment. Retrospective AKI scoring is still not executed because this run has neither a probability-complete EHR-trained target policy nor its policy-specific post-training overlap/ESS receipt. Thus no estimator is transported by family name, no retrospective EHR OPE value is reported, and all 16,128 reference and task-matched tuple metric/disposition rows are released.

5.4 RQ4: Cross-cohort evaluability

The common scaffold produces time-ordered transition tasks in all six cohorts, but higher evaluation layers differ. Respiratory and shock retain rich repeated actions, post-action reward proxies, P/R/T components, and known-value policy matrices. AF/flutter has a supported $K = 25$ transition action but fails strict action-reward ordering and remains world-model-only. AKI and heart failure require separate decision contracts rather than promotion of their compact transition actions: daily absorbing RRT initiation and six-hour loop-diuretic by vaso/inotrope, respectively. Both support known-value model-based/model-free comparison; only AKI yields task-matched semi-synthetic OPE candidates, and neither reaches retrospective EHR value scoring. The benchmark therefore reports a six-task world-model surface, four known-value policy surfaces, one world-model-only task, and zero retrospective policy-value surfaces.

6 PREDICTION-TO-POLICY FAILURE DECOMPOSITION

Prediction quality is neither ignored nor treated as policy performance. The machine-readable bridge links one-step and recursive errors, reward error, termination calibration, factual action information, support error, model-exploitation gap, known-value

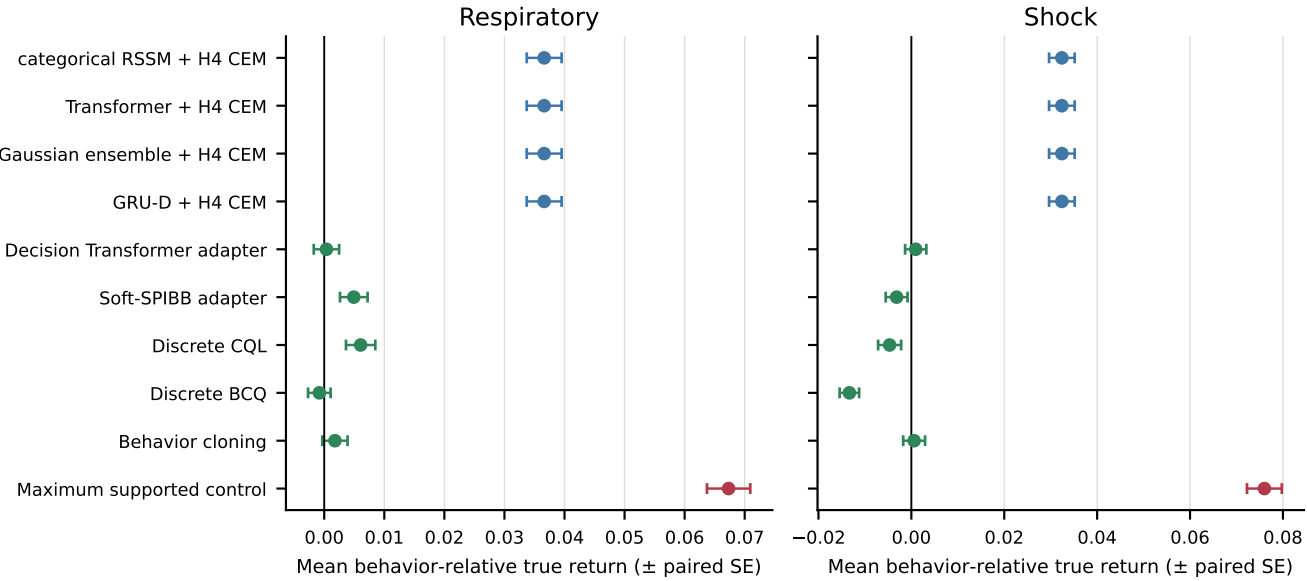


Figure 1: Policy-level paired behavior-relative return in moderate primary known-value regimes. Error bars are mean common-random-number paired Monte Carlo SE, not across-method intervals. H4 uses the identical support-constrained planner contract for all four world models. Raw reward values are not compared across tasks.

return, and retrospective scoring status for every core world model. Four patterns are important.

First, one-step rankings do not persist under recursive rollout: the Transformer-to-ensemble reversal occurs in five cohorts and Transformer-to-LOCF in heart failure; among the four learned world models, the Transformer-to-ensemble reversal occurs in all six. Second, cohort-level uncertainty shows that recursive CRPS and recursive calibration select different methods. Third, mean absolute learned-model-versus-true-return gaps are 1.079 across 72 respiratory planner rows and 2.344 across 72 shock rows, so simulated value cannot substitute for known truth. Finally, an extreme-action control leads every moderate known-value task, but the preferred extreme flips for AKI; together with H4 ties or narrow planner ranges, this shows that a positive planner return does not establish useful world-model discrimination. Exploratory prediction-policy correlations share environments and learned models and are not treated as independent evidence.

The resulting benchmark logic is compositional: a row may remain valid for P/R/T forecasting while failing planning discrimination, and a policy may remain valid for support/collapse analysis while failing value authorization. This prevents an upper-layer failure from erasing lower-layer evidence or a lower-layer success from silently authorizing a higher-layer claim.

7 REPRODUCIBILITY AND ACCESS

The reader package includes a benchmark card; task, feature, action, reward, planner, model, and estimator cards; sanitized policy-level and tuple-level result tables; a number-to-source audit; immutable source hashes; environment manifest; build

commands; and credentialed reconstruction instructions. It exposes the 54 layer-specific method identities, all 2,448 known-value policy rows, all 16,128 OPE tuple metric/disposition rows, horizon-resolved transition results, cohort-specific uncertainty leaders, and gate thresholds rather than only headline aggregates. Code, contracts, synthetic fixtures, aggregate evidence, and clean-room commands are provided in the anonymous artifact at <https://anonymous.4open.science/r/ehrdyn-icu-65FB>. The source data are MIMIC-IV v3.1 [5]; access remains governed by PhysioNet credentialing and its data-use agreement. Raw EHR rows, identifiers, exact timestamps, trajectories, role-membership files, row-level predictions, tensors, and checkpoints are not redistributed.

Authorized users rebuild task arrays locally, fit all data-dependent transforms on development training roles, run development validation, execute pure and aggregate-calibrated known-value evaluation, and regenerate the aggregate synthesis and PDF. Unrestricted users can validate the packaged contracts and aggregate hashes and run the known-value smoke test without MIMIC-IV. Independent credentialed numerical reconstruction remains pending and is not conflated with synthetic clean-room execution.

8 LIMITATIONS, DATA GOVERNANCE, AND SCOPE

The EHR findings are development benchmark evidence: no genuinely untouched confirmatory role is available. World-model leader comparisons lack subject-cluster intervals, and the derived Gaussian ensemble contributes one row per task. Recursive

evaluation predicts values under a fixed missingness convention rather than modeling a generative observation process. The four-hour action is a recorded interval exposure, not guaranteed to be assignable at the bin start. Sepsis is published-compatible with local parameterization rather than an exact reproduction; its reference terminal outcome is unavailable. AKI and heart-failure policy comparisons use separately constructed known-value tasks rather than the original EHR transition actions, and AF remains world-model-only.

The known-value mechanisms are transparent but largely monotone and do not test heterogeneous or non-monotone optima. AKI has task-matched semi-synthetic OPE approvals, but no task supplies the trained target-policy and post-training gates needed for retrospective scoring; the benchmark therefore does not rank policies on EHR data. MIMIC-IV is restricted, so the artifact shares code, contracts, synthetic fixtures, and aggregate results rather than row-level data. These are evidence boundaries of the technical benchmark.

9 CONCLUSION

We present a common decision-evaluation scaffold that connects EHR world-model learning, recursive uncertainty, planning, model-free RL, known-value truth, and OPE authorization. Its main empirical finding is not a winner: one-step, recursive, probabilistic, planning, and estimator layers produce different rankings and failure modes. By publishing exact contracts, granular rows, fixed controls, and non-execution rules, the benchmark makes those differences auditable while preserving valid lower-layer results. Retrospective EHR policy ranking remains unavailable until an exact evaluator contract passes the frozen bridge.

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A AGGREGATE RESULT INTERFACES

The aggregate release is intentionally row-granular without exposing patient rows. Table 6 lists the primary machine-readable interfaces used to regenerate the manuscript. Every file is generated from frozen sources by one package builder and included in the hash manifest.

B APPROVED EXACT-FINITE OPE SCOPE

All 32 approved tuples use the moderate exact-finite regime, horizon one, and masked support. Counts by estimator are: CWPDIS 3, DR 3, IS 5, PDIS 5, WDR 7, WIS 3, WPDIS 3, and support-restricted WPDIS 3. Denominator counts are exact behavior 16, stronger cross-fit 8, deliberately misspecified behavior 6, and the paper-configuration LSTM 2; 20 tuples use clipping 20 and 12 use no clipping. This inventory is descriptive of the exact-finite scope only. No passing identity is transported to the aggregate-calibrated or retrospective surfaces.

The task-matched evaluator separately approves 228 exact-finite and 236 semi-synthetic AKI tuples and zero heart-failure tuples. The AKI semi-synthetic approvals span weak, moderate, and delayed regimes; horizons one, two, and three; and all eight estimator families represented in the reference matrix. They remain environment-specific tuple approvals, not an estimator-family endorsement or retrospective authorization.

C RECONSTRUCTION BOUNDARY

The credentialed reconstruction command materializes arrays locally from MIMIC-IV, fits train-only transforms, runs development validation, and exports only aggregate tables. The unrestricted smoke-test path instead uses schema-compatible synthetic fixtures and validates action masks, planner horizons, categorical-CEM iterations, common random numbers, OPE formulas, table schemas, hashes, and privacy scans. An external run that reproduces only the smoke-test path does not constitute credentialed numerical parity.

Table 6: Reader-facing aggregate interfaces.

File	Unit and purpose	Rows
cohort_scale.csv	Task-level subject, transition, action, support, and known-value scenario counts	6
baseline_method_inventory.csv	Complete layer-specific transition, uncertainty, policy, and OPE method inventory	54
baseline_surface_atlas.csv	Cross-layer inventory, scope, and point-leader patterns used in Table 2	6
all_baseline_transition_leaders.csv	Cohort-specific one-step and final-horizon leaders across all six transition methods	12
factual_action_information_primary.csv	Fixed primary action-information estimand and interval counts	6
world_model_planner_bridge.csv	World model $\times$ cohort P/R/T, prediction, uncertainty, action, and known-value bridge	24
cohort_uncertainty_leaders.csv	Cohort-level CRPS, MACE, and recursive coverage ranges across five probabilistic methods	6
world_model_uncertainty_summary.csv	Task-balanced core-world-model probabilistic metrics	8
world_model_recursive_horizon_metrics.csv	Horizon-resolved logged-action recursive RMSE/MAE and observed-cell counts	924
known_value_policy_level.csv	Every fixed policy/planner row in the moderate primary regime	68
known_value_reference_full_matrix.csv	Respiratory/shock method $\times$ seed $\times$ response $\times$ reward rows	1,632
known_value_task_extension_full_matrix.csv	AKI/HF method $\times$ seed $\times$ response rows	816
ope_gate_breakdown.csv	Gate pass counts for both known-value tiers	2
ope_reference_all_tuple_metrics.csv	Reference evaluator metrics and all-nine dispositions	6,912
ope_task_matched_all_tuple_metrics.csv	AKI/HF task-matched metrics and all-nine dispositions	9,216
number-audit.csv	Manuscript token to source, estimand, split, and uncertainty mapping	variable